Multiscale CNN Based on Component Analysis for SAR ATR

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***Abstract*— This article proposes a multiscale convolutional neural network (CNN) based on component analysis (CA-MCNN) for synthetic aperture radar (SAR) automatic target recognition (ATR). The component information of a target is robust to the local variations of the target, which is not made the best of by traditional CNN-based methods. For learning the component information, we use the attributed scattering centers (ASCs) extracted from the target echoes as the components of the target for SAR ATR, which divides the SAR target according to the geometric scattering types of ASCs and can not only make the division results more robust but also accurately characterize the electromagnetic scattering characteristics of the target. Since the global information provided by the whole image is also important for SAR ATR, CA-MCNN combines the global information with component information to learn a more efficient and robust target feature representation. In addition, considering that the feature maps of the shallower layer in CNN focus on local and fine-grained information while the feature maps in the deeper layer focus on global and coarse-grained information, we fuse the multiscale feature maps obtained from different layers to enhance the feature description ability. Extensive experiments conducted on the moving and stationary target acquisition and recognition (MSTAR) data set prove the superior performance of CA-MCNN.**

***Index Terms*— Attributed scattering centers (ASCs), automatic target recognition (ATR), component information, convolutional neural networks (CNNs), global information, multiscale, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) is an active ground observation system, which uses the principle of synthetic aperture to realize high-resolution microwave imaging. With a wealth of irreplaceable characteristics such as high-resolution, all-time and all-weather, SAR has been widely used in many applications both in civil and military fields. However, SAR images have different imaging mechanism and unique speckle noise compared with optical images, which makes the interpretation of SAR images more difficult than that of optical images [1]. Therefore, how to quickly and accurately interpret SAR images to obtain valuable information has received widespread attention. As an important part of SAR image interpretation, SAR target recognition can help humans get target category information from SAR images [2]. With the development of SAR imaging technology, how to realize SAR automatic target recognition (ATR) accurately and effectively has significant application value both in civil and military fields.

For target recognition, feature extraction is critical to the recognition performance. How to extract effective features has become the core problem of the task. In the existing work, many SAR target recognition methods based on shallow features and classifiers are proposed. In these methods, the feature extraction methods of SAR images can be divided into two categories: one is to find the key points in the SAR image first, and then describe these key points in vectors, such as histogram of oriented gradients (HOG) [3], local binary patterns (LBP) [4], and scale invariant feature transform (SIFT) [5]; the other is to directly map SAR images to feature vectors through linear or nonlinear transformations, such as fisher linear discriminant analysis (LDA) [6], principal component analysis (PCA) [6], and independent component analysis (ICA) [7]. However, such artificially designed shallow features of SAR image are generally based on the long-term knowledge accumulation of experts, which makes the features have limited representation capabilities for recognition. In addition, SAR target recognition methods based on shallow features and classifiers need to extract different suitable sample features for different data sets based on artificial experience, which are highly targeted and have poor generalization.

With the improvement of hardware conditions and the emergence of large-scale data sets, deep learning has become the focus of research in various fields. The excellent feature self-learning ability has made deep learning widely concerned. Inspired by the cognitive mechanism of the human brain, deep learning builds a multilevel model structure to map the original data into the feature space, where the targets are more conducive to be classified correctly, through the multilayer nonlinear transformation. As a representative algorithm of deep learning, convolutional neural network (CNN) uses a unified framework to achieve the feature extraction and classification, which effectively solves the difficulty of designing manual features in the past. Several studies [8]–[10] confirmed that the convolution operation can make the learned deep features tend to be more discriminative for different classes of targets. At present, there are many CNN-based methods for SAR ATR. Ding *et al.* [11] introduced three types of data augmentation operation to augment the training data,

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including translation, speckle noising, and pose synthesis. The method in [11] can enhance the completeness of training set and alleviate the overfitting of CNN caused by insufficient training samples. Chen *et al.* [1] propose a new network for SAR ATR, which only includes convolution layers and has no fully connected layers. The network in [1] can also solve the problem to a certain extent that CNN is easy to overfit without sufficient training data. Li *et al.* [12] decompose the entire CNN into a convolutional self-encoding network and a shallow neural network, thereby improving the training efficiency of CNN. Leonan *et al.* [13] use VGG-16 and VGG-19 to realize the recognition of oil rigs on Sentinel-1 SAR images.

The above-mentioned CNN-based methods for SAR ATR directly process the whole image, which do not make the best of the component information of the target. For a target, the component information is robust to the local variations, which is also useful for SAR ATR. For example, in moving and stationary target acquisition and recognition (MSTAR) data set, there are some variants that only exist in the test data. It is difficult to properly classify these variants using only the global information of the whole image. However, compared with the original type, only a few components of these variants have changed, and most of the components are unchanged. Therefore, it is easier to correctly classify the variants using the unchanged components. In this situation, introducing component information of the target into CNN can improve the accuracy of variant recognition. Therefore, we hope to propose a method to effectively use the component information of the target to further improve the performance of SAR ATR.

For learning the component information, we consider to divide the SAR target into multiple components first. However, due to the different imaging mechanisms, SAR images have unique electromagnetic scattering characteristics compared with optical images. In the high-frequency region, the radar backscattering of a distributed object can be approximated using the sum of the responses from various individual scattering centers [14]. To effectively characterize the electromagnetic scattering characteristics of scattering centers, researchers proposed the attributed scattering center (ASC) model [14], which characterizes the dependence of the target's backscattering response on frequency and azimuth. Its model parameters contain rich physical and geometric attributes, which can more accurately characterize the electromagnetic scattering characteristics of radar targets in the high-frequency region. In addition, the ASCs of an SAR target can reflect the real geometrical structure of target components, such as dihedral, sphere, and so on, and be more robust to target variants and environmental changes. Therefore, we consider to use the ASCs extracted from the target as the components of the target to learn the component information. This method divides the SAR target according to the geometric scattering type of ASCs, such as dihedral, sphere, and so on, which can not only make the division results more robust but also accurately characterize the electromagnetic scattering characteristics of the target.

Besides the component information of the target, the global information provided by the whole image contains the overall information and the shadow information of the target, which is also very important for SAR ATR. Therefore, a multiscale CNN based on component analysis (CA-MCNN) for SAR ATR is proposed in this article, which combines the global information from the whole SAR image with component information corresponding to ASC components to learn a more efficient and robust target feature representation. There are three main modules in our method: backbone module, component feature maps module, and feature fusion module. The backbone module contains some convolution layers and the fusion of multiscale feature maps, which aims at obtaining the more descriptive global feature maps of the input SAR image. Then, the component feature maps module obtains the component feature maps through multiplying the global feature maps and the component binary maps obtained via the ASC model. Next, the feature fusion module converts the global feature maps into global feature representation through global average pooling (GAP) [15], and converts component feature maps into component feature representations through GAP and ASC components fusion, and then fuses them to obtain a more efficient and robust feature representation. Finally, the fused feature representation is fed into the softmax layer to classify the input image.

Our contributions can be summarized as follows.

1) Using the extracted ASCs as the target components to learn the component feature maps. Our proposed method divides the SAR target according to the geometric scattering type of ASCs, such as dihedral, sphere, and so on, which can not only make the division results more robust but also accurately characterize the electromagnetic scattering characteristics of the target. In addition, since the obtained component feature maps correspond to the types of ASCs, they are more interpretable compared with the feature maps obtained by the traditional CNN.
2) Combining global information from the whole SAR image with component information corresponding to ASCs of the target within CNN framework. The global information provided by the whole image contains the overall information and the shadow information of the target, while the component information is robust to the local variations of the target, both of which are important for SAR ATR. Therefore, combining the global information with component information within CNN framework can construct a more efficient and robust target feature representation model.
3) Using the fusion of multiscale feature maps to enhance the description ability of global feature maps. In CNN, the receptive field of different layers becomes larger as the layer becomes deeper. The feature maps obtained from the shallower layer focus on local and fine-grained information while the feature maps obtained from the deeper layer focus on global and coarse-grained information. Therefore, fusing the multiscale feature maps of different layers can enhance the description ability of the global feature maps.

The rest of this article is organized as follows. First, we introduce the basic theory of CNN, ASC model, and ASCs extraction in Section II. Section III explains the details of the proposed CA-MCNN. Then extensive experiments on MSTAR data set and related analysis are shown in Section IV. Finally, Section V conclude the whole article.

II. Preliminary Knowledge

In this section, we introduce the preliminary knowledge related to the proposed method to better understand the work of the article, including the basic theory of CNN, ASC model, and ASCs extraction.

A. Convolutional Neural Network

As a representative algorithm of deep learning, CNN has become the most popular tool in image interpretation [8]–[10]. Because of the deep structure, CNN can automatically dig the complex structure of data by integrating the simple features extracted from the previous layers into layer-by-layer abstract features that are more suitable for classification, which effectively solves the difficulty of designing manual features in the past.

CNN uses an unified framework to achieve the feature extraction and classification. The feature extraction part in CNN mainly includes the convolution layer and pooling layer. In the convolution layer, the previous layer's input feature maps $O_i^{(l-1)}(i = 1, \ldots, I)$ are connected to all the output feature maps $O_j^{(l)}(j = 1, \ldots, J)$, where $O_i^{(i-1)}(p, q)$ and $O_j^{(l)}(p, q)$ are the unit of the ith input feature map and jth output feature map at position (p, q), respectively. Each unit in the output feature map can be calculated as follows [1]:

$$V_j^{(l)}(p, q) = f\left(\sum_{i=1}^{I}\sum_{u,v=0}^{F-1} k_{ji}^{(l)}(u, v) \cdot O_i^{(l-1)}(p-u, q-v) + b_j^{(l)}\right) \quad (1)$$

where $k_{ji}^{(l)}(u, v)$ is the convolutional kernel of size $F \times F$; $f(\cdot)$ denotes the nonlinear activation function and $b_j^{(l)}$ represents the bias.

There is often a pooling operation after the convolution layer. Pooling operations can not only reduce the computational cost of the network, but also make the network more robust to translation, rotation, and scaling. The pooling methods commonly used in CNN are either selecting the maximum or average in the pooled area with the size of $h \times w$. For example, the max pooling is defined as follows [1]:

$$O_i^{(l+1)}(p, q) = \max_{u,v-0,\ldots,G-1}\left(O_i^{(l)}(p \cdot s + u, q \cdot s + v)\right) \quad (2)$$

where G is the pooling size, and s is the stride, which determines the intervals between neighbor pooling windows.

The classification part in CNN uses the softmax function to determine the target label. The final output is a vector, where the value represents the probability that the input image belongs to each category. The softmax function is defined as follows [16]:

$$p_i = \frac{\exp\left(O_i^L\right)}{\sum_{j=1}^{K}\exp\left(O_j^L\right)} \quad (3)$$

where $O_j^{(L)}$ represents the weighted sum of inputs to the jth unit on the output layer, and K represents the number of units on the output layer. The target label will be determined as the class with the maximum probability.

B. Attributed Scattering Center Model

Using the sum of the responses from various individual scattering centers can approximate a distributed object's radar backscattering in the high-frequency region, which can be described as follows [14]:

$$E(f, \varphi; \theta) = \sum_{i=1}^{K} E_i(f, \varphi; \theta_i) \quad (4)$$

where f represents the frequency; φ denotes the aspect angle; the number of the ASCs in the object is denoted as K; the ASCs' parameter set is represented by $\theta = \{\theta_i\} = [A_i, \alpha_i, x_i, y_i, L_i, \bar{\varphi}_i, \gamma_i](i = 1, 2, \ldots, K)$. For a single ASC, we can use the ASC model to describe its backscattering field as follows [14]:

$$\begin{aligned} E_i(f, \varphi; \theta_i) &= A_i \cdot \left(j\frac{f}{f_c}\right)^{\alpha_i} \cdot \exp\left(\frac{-j4\pi f}{c}(x_i \cos\varphi + y_i \sin\varphi)\right) \\ &\quad \cdot \sin c\left(\frac{2\pi f}{c} L_i \sin(\varphi - \bar{\varphi}_i)\right) \cdot \exp(-2\pi f \gamma_i \sin\varphi) \end{aligned} \quad (5)$$

where c represents the propagation velocity of the electromagnetic wave; f_c denotes radar center frequency; for the ith ASC, the complex amplitude is represented by A_i; the frequency dependence is denoted by α_i; the position coordinates of the scattering center in the range dimension and azimuth dimension are denoted by x_i and y_i, respectively; the length of the distributed ASC is represented by L_i and the orientation of the distributed ASC is represented by $\overline{\varphi_i}$; the aspect dependence of the localized ASC is denoted by γ_i.

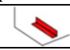
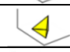
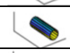
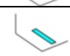
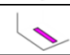
The ASC model describes the dependence of the target's backscattering response on frequency and azimuth using a set of parameters. These parameters describe the physical characteristics of target scattering centers, including position, shape, orientation (pose), and relative amplitude [14]. Among these parameters, two of them are selected to differentiate eight geometric scattering types of ASCs, which are listed in Table I. As we can see in Table I, different colors and shapes are used to represent different types of ASCs. The colors are used to enhance the differences between different types of ASCs, for example, if we only use shapes to represent different types of ASCs, the edge broadside and edge diffraction cannot be distinguished.

C. Attributed Scattering Centers Extraction

ASC parameter estimation is not only a high-dimensional problem, but also a nonlinear and nonconvex problem. Currently, to estimate ASC parameters, many algorithms based on image domain processing and algorithms based on frequency domain processing have been proposed. Most existing algorithms based on image domain processing extract ASCs from the radar backscatters based on image segmentation in

TABLE I

GEOMETRIC SCATTERING TYPES INFERENCE FROM FREQUENCY DEPENDENCE AND LENGTH

Geometric scattering type	Frequency dependence (α)	Length (L)	Iconic representation
Dihedral	1	> 0	
Trihedral	1	0	
Cylinder	0.5	> 0	
Top Hat	0.5	0	
Sphere	0	0	
Edge Broadside (EB)	0	> 0	
Edge Diffraction (ED)	-0.5	> 0	
Corner Diffraction (CD)	-1	0	

the image domain [17]. However, if the image segmentation cannot separate each ASC correctly, these algorithms may not get accurate estimation results. Although the algorithms based on frequency domain processing do not require image segmentation, their computational complexity and storage requirement are usually high.

In our article, we use the ASCs of different geometric scattering types to represent components of different types, that is, the ASCs whose geometric scattering types are dihedral are the first type of components, the ASCs whose geometric scattering types are cylinder are the second type of components, and so on. Since we use ASCs to obtain component binary maps, which reflect some information of ASCs extracted from target, including location, length, and orientation, and can be used to describe the component information corresponding to ASC components extracted from the target, the accurate ASC extraction result is important for the final recognition result. For obtaining the accurate component binary maps, the method in [18] is adopted to extract ASCs from the input image, which is a novel ASC extraction algorithm based on image domain processing. This method first converts SAR measurements to sparse representations in the image domain. Then, a newtonlized orthogonal matching pursuit (NOMP) algorithm is adopted to estimate the ASC model parameters. The ASCs extraction algorithm in [18] has good accuracy and fast calculation speed, which is summarized as Algorithm 1.

Algorithm 1 ASCs Extraction Algorithm Proposed in [18]

Input: SAR image $S(x, y)$, dictionary $\Phi = \left\{\hat{D}(x, y; \boldsymbol{\theta}_q)|\theta_q \in \Theta\right\}$, where $\hat{D}(x, y; \boldsymbol{\theta}_q)$ is the normalized ASC image, $\boldsymbol{\theta}_q$ is the ASC parameters set of q-th ASC and Θ contains all possible values of ASC parameters.

Output: $\left(\boldsymbol{\theta}_{q,opt,\sigma_q^{Q,opt}}\right)$, q = 1, 2, ..., Q

Initialization: residual image $R(x, y) := S(x, y)$, $\Phi_{Gen} = \emptyset$, $i := 1$

while the stop criterion is not met. **do**

1. Coarsely estimating the parameter of the i-th ASC, denoted by θ_i, by selecting the most matched atom $\hat{D}(x, y; \boldsymbol{\theta}_q) \in \Phi$ for $R(x, y)$, which can be done by evaluating the inner product of the residual image and every atom in Φ, as given by:
$$\boldsymbol{\theta}_i = \arg\max_{\theta_q} \left| \int\int R(x, y)\hat{D}^*(x, y; \boldsymbol{\theta}_q)dxdy \right|$$
2. Refining the estimation of the continuous parameters of the i-th ASC (i.e., $x, y, L, \bar{\varphi}$ and γ) by taking $\boldsymbol{\theta}_i$ as the starting point and running Newton's method:
$$\boldsymbol{\theta}_{i,opt} = \arg\max_{\boldsymbol{\theta}\backslash\{A\}} \left| \int\int R(x, y)\hat{D}^*(x, y; \boldsymbol{\theta})dxdy \right|$$
Generating new atom according to $\boldsymbol{\theta}_{i,opt}$, and put the new atom into generated atoms collection Φ_{Gen}: $\Phi_{Gen} := \Phi_{Gen} \cup \left\{D(x, y; \boldsymbol{\theta}_{i,opt})\right\}$.
3. Using atoms in Φ_{Gen} to approximate the original image $S(x, y)$. The least-square estimation is carried out to estimate the coefficients $\boldsymbol{\sigma}^{i,opt}$ of the i selected atoms, as given by:
$$\boldsymbol{\sigma}^{i,opt} = \arg\min_{\sigma^i} \left\| S(x, y) - \sum_{q=1}^{i} \sigma_q^i D\big(x, y; \boldsymbol{\theta}_{q,opt}\big) \right\|_2^2$$
4. Updating the residual image by canceling the i selected atoms, i.e., $R(x, y) := S(x, y) - \sum_{q=1}^{i} \sigma_q^{i,opt} D(x, y; \boldsymbol{\theta}_{q,opt})$. $i := i + 1$.

end while

III. METHODOLOGY

In this section, a detailed introduction of the proposed CA-MCNN will be presented. We first describe the overall framework of CA-MCNN. Then, we elaborate the three main modules of the proposed multiscale CNN model, respectively.

A. Overall Framework

The overall framework of CA-MCNN is shown in Fig. 1, which contains three main modules, i.e., backbone module, component feature maps module, and feature fusion module. The input of the backbone module is a real-valued SAR image, which is obtained by performing a modulo operation on the complex SAR data. The output of the backbone module is global feature maps describing the global information of the whole image. In addition, the backbone module contains the fusion of multiscale feature maps, which can enhance the description ability of global feature maps. Through the ASC model, the ASCs of the input complex SAR data can be extracted, which can be used to obtain component binary maps. Then, the global feature maps and component binary maps are input into the component feature maps module to obtain the component feature maps that describe the component information corresponding to ASC components. The feature fusion module aims at fusing the global information and component information to obtain a more efficient and robust feature representation of the input image. Finally, the obtained feature representation is fed into the softmax layer

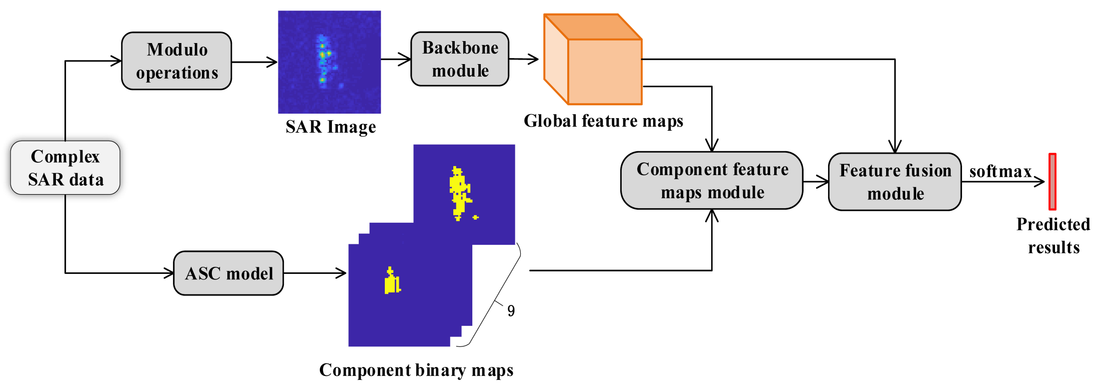


Fig. 1. Overall framework of CA-MCNN.

to classify the input image. The entire network is trained in an end-to-end way and the stochastic gradient descent method is used to update the network parameters. Besides, the loss function of CA-MCNN is the cross-entropy loss function.

B. Backbone Module

The backbone module of CA-MCNN aims at obtaining the more descriptive global feature maps of the input image. At present, many excellent convolutional network architectures, such as VGGNet [19] and ResNet [10], have been widely used in many fields. However, these above-mentioned networks have a larger number of parameters and require large amounts of labeled samples to estimate them. In the real scenarios, collecting large amounts of SAR images with labels is very costly and difficult. Therefore, we built a simple CNN architecture as the backbone module of our method to extract features of the input SAR image, which has much fewer parameters and can achieve approximate performance meanwhile.

As shown in Fig. 2, there are only four convolutional layers in the backbone module. The resolution of input image is 64×64 pixels, and the resolution of the global feature maps obtained by the backbone module is also 64×64. Such a network structure can not only fully learn the features of the input image, but also make the resolution of the obtained global feature maps not too low to obscure the component information of the target. In the backbone module, the first three convolutional layers are used to extract features from the input SAR image. Considering that the training SAR images with labels are difficult to be collected, we hope the backbone module of our method can obtain the more descriptive global feature maps. Therefore, we perform the multiscale fusion to enhance the description of the global feature maps. The detailed operation of multiscale fusion is concatenating the feature maps obtained from the first three layers along the channel axis and using the fourth convolutional layer to learn proper weights to better combine the information extracted by different convolutional layers. After the backbone module, the more descriptive global feature maps of the input SAR image can be obtained.

C. Component Feature Maps Module

We use the ASCs of different geometric scattering types to represent components of different types. Therefore, there are eight types of components in our article, including dihedral, sphere, and so on. The component feature maps module of CA-MCNN aims at obtaining the component feature maps, which describe the component information corresponding to ASC components extracted from the target. Fig. 3 shows the detailed operation of this module. We first use the method in Section II-C to extract the ASCs of the input complex SAR data. Then, the extracted parameters of ASCs are used to obtain the component reconstruction maps, which include one reconstruction map corresponding to all components and eight reconstruction maps corresponding to eight types of components. As shown in Fig. 4(a) is an original SAR image that contains target, shadow, and noise, which reflects the global information of the image; Fig. 4(c)–(k) are component reconstruction maps corresponding to ASC components extracted from the target in Fig. 4(a), which reflect the component information and only retain the characteristics of the component regions without considering other regions. Compared with Fig. 4(a), the reconstruction map that corresponds to all ASCs in Fig. 4(c) pays more attention to the areas of the target with higher scattering intensity and contains almost no noise. Compared with the reconstruction maps corresponding to eight geometric scattering types of ASCs in Fig. 4(d)–(k), the reconstruction map in Fig. 4(c) is the combination of them and can reflect the overall situation of all extracted ASC components. Next, we convert all component reconstruction maps into binary maps by setting a threshold, which can be called as component binary maps. The component binary maps include one binary map corresponding to all components and eight binary maps corresponding to eight types of components.

As shown in Fig. 3, after obtaining the component binary maps, we multiply them with global feature maps to obtain the component feature maps. The component feature maps include a set of feature maps corresponding to all components and eight sets of feature maps corresponding to eight types of components, which describe the component information corresponding to ASC components extracted from the input complex SAR data.

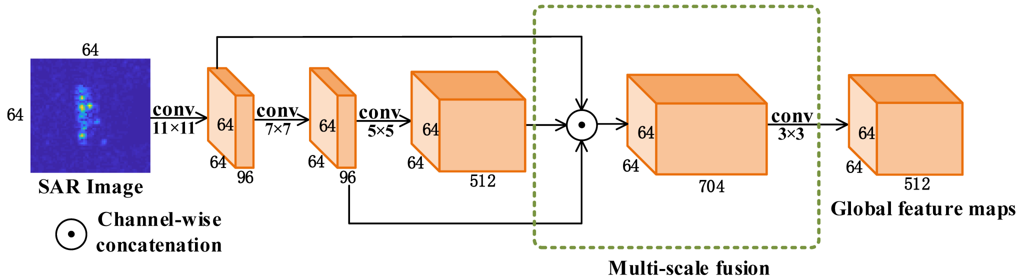


Fig. 2. Backbone module of CA-MCNN. "Conv" represents the convolution operation. The filter size of the convolution operation is represented as "$m \times m$."

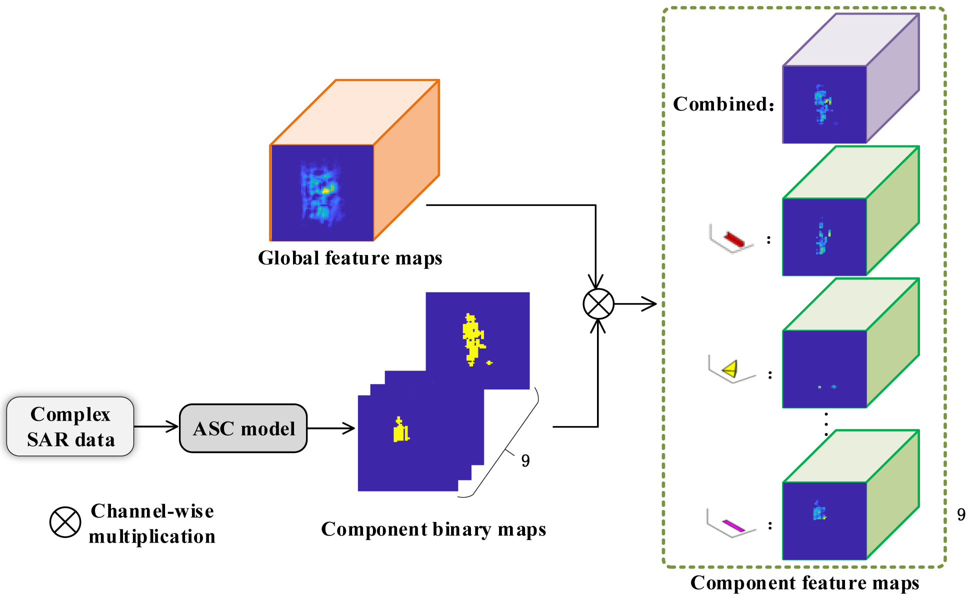


Fig. 3. Component feature maps module of CA-MCNN. The component feature maps include a set of feature maps corresponding to all components and eight sets of feature maps corresponding to eight types of components extracted from the input complex SAR data.

D. Feature Fusion Module

The feature fusion module aims at fusing the global information and component information of the input image to obtain a more efficient and robust feature representation of the input image. It can be seen from Fig. 5, before the global-component information fusion, we need to obtain the global feature representation, $S1$, the combined components feature representation, $S2$, and the local component feature representation, $S3$, first. The global feature representation, $S1$, of the input image is obtained by performing GAP on global feature maps, which reflects the global information of the whole image and is important for SAR ATR. The combined components feature representation, $S2$, of the input image is obtained by performing GAP on component feature maps corresponding to all ASCs, which not only reflects the overall situation of all extracted ASC components, but also reflects the areas of the target with higher scattering intensity. The local component feature representation, $S3$, of the input image is obtained by GAP and ASC components fusion, which describes the component information corresponding to the eight types of ASC components extracted from the target. Then, the global-component information fusion is to concatenate $S1$–$S3$ to obtain one vector, and then feeds it into the full-connection layer to produce the final feature representation of the input image, which is more effective and robust. The following is a detailed introduction of ASC components fusion.

As shown in Fig. 5, before ASC components fusion, we need to obtain local component feature vectors first, which are obtained by performing GAP on the eight sets of component feature maps corresponding to eight types of components, respectively. Considering that not each local component plays the same important role in the target feature representation, and to reduce the dimension of the feature representation, we perform ASC components fusion on the eight local component feature vectors and compressed them into one vector, $S3$.

To achieve ASC components fusion, this article introduces several instantiations of functions. As shown in Fig. 5, we perform one of the following functions on the corresponding positions of the eight local component feature vectors, thereby integrating them into one vector.

1) Statistical Functions: To compress the eight local component feature vectors into one vector, we can use statistical functions on the corresponding positions of the eight vectors.

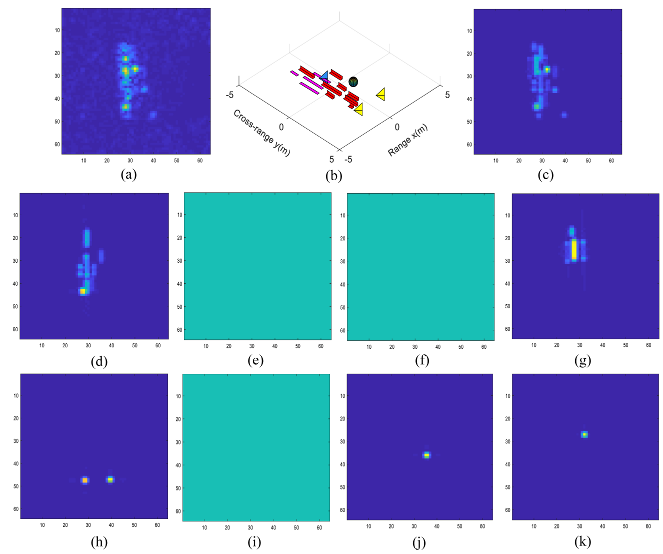


Fig. 4. Component reconstruction maps of a BMP2. (a) Original SAR image. (b) ASC schematic map. (c) Reconstruction map corresponding to all ASCs. (d) Dihedral reconstruction map. (e) Cylinder reconstruction map. (f) Edge broadside reconstruction map. (g) Edge Diffraction reconstruction map. (h) Trihedral reconstruction map. (i) Top hat reconstruction map. (j) Sphere reconstruction map. (k) Corner diffraction reconstruction map. The ASC schematic map reflects all geometric scattering types of ASCs in the original SAR image. As shown in (b), there are only five geometric scattering types of ASCs in (a). Therefore, if a geometric scattering type of ASC does not exist in an SAR image, the component reconstruction map corresponding to this ASC type will be set to 0.

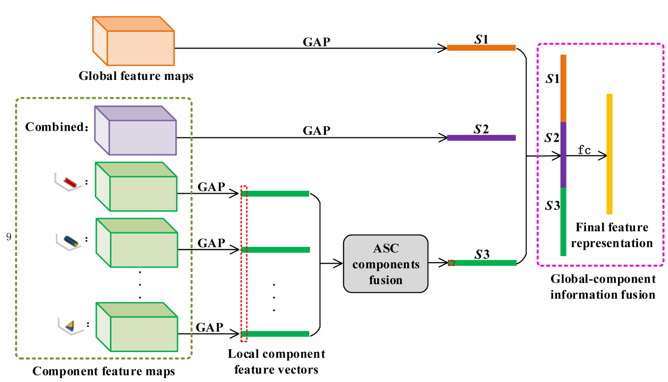


Fig. 5. Feature fusion module of CA-MCNN. "GAP" represents the global average pooling operation, "$S1$" represents the global feature representation, "$S2$" represents the combined components feature representation, "$S3$" represents the local component feature representation, "fc" indicates the fully connected operation.

Considering the representativeness and the computational cost, two statistical functions are selected: $\max(\cdot)$ and $\text{mean}(\cdot)$. Section IV-D will show the comparison of all selected functions.

2) Joint Functions: Based on the two selected statistical functions, we apply two ways to joint them as follows:

$$G_1(\cdot) = \max(\cdot) + \text{mean}(\cdot) \tag{6}$$
$$G_2(\cdot) = 1\_1C(\text{cat}(\max(\cdot), \text{mean}(\cdot))) \tag{7}$$

where $\max(\cdot)$ and $\text{mean}(\cdot)$ are applied on the corresponding positions of the eight local component feature vectors, 1_1C represents 1×1 convolutional layer and cat denotes concatenation on the channel dimension. In (7), the 1×1 convolutional layer is used to learn weights of information extracted by different statistical functions, so as to realize the fusion of various information.

TABLE II

DETAILED INFORMATION OF THREE-TARGET DATA

Dataset	BMP2			BTR70	T72		
	C21	9566	9563	C71	132	S7	812
Training (17°)	233	0	0	233	232	0	0
Test (15°)	196	196	195	196	196	191	195

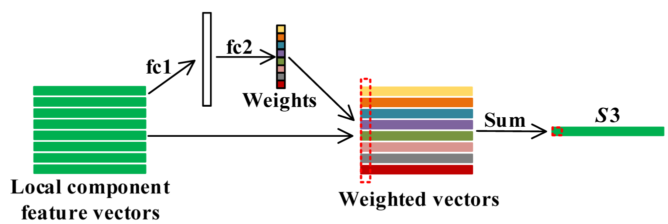


Fig. 6. Structure of the ASC components fusion using attention. "fc1" and "fc2" indicate the fully connected operation, "sum" represents the summation of corresponding positions of the eight weighted vectors.

3) Attention: Considering that visual attention has been widely used in lots of tasks [20], [21], we use it to achieve the ASC components fusion. As shown in Fig. 6, we use the squeeze-and-excitation (SE) network [22] to learn weights of the eight local component feature vectors first, and then we sum the eight weighted vectors to obtain one vector.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Experimental Data Description

In this article, the MSTAR data set, which is collected using the 10-GHz SAR sensor in 0.3 × 0.3 m resolution, is used in experiments. There are ten different ground military targets in the MSTAR data set, including BMP2 (tank), BTR70 (armored vehicle), T72 (tank), BTR60 (armored vehicle), 2S1 (cannon), BRDM (truck), D7 (bulldozer), T62 (tank), ZIL131 (truck), and ZSU234 (cannon). Among them, BMP2 and T72 have variants in test stage. Fig. 7 shows some examples of the ten different ground targets, including SAR images and their corresponding optical images.

In the MSTAR data set, the depression angles of samples for each target category are 15° and 17°, and the aspect angles cover from 0° to 360°. Referring to the existing literatures, this article focuses on two experimental scenarios: three-target data SAR target recognition and ten-target data SAR target recognition. The specific details of experimental data setting on the above-mentioned two experimental scenarios are listed in Tables II and III, respectively. It can be seen from Table II, the three-target data include T72, BMP2, and BTR70, and BMP2 and T72 both include three different serial numbers. The training data only include one serial number of each type, while the test data include all serial numbers of each type. In addition, the depression angles of training data and test data

TABLE III
DETAILED INFORMATION OF TEN-TARGET DATA

Dataset	BMP2	BTR70	T72	BTR60	2S1	BRDM2	D7	T62	ZIL131	ZSU234
Training (17°)	233 (C21)	233	232 (132)	255	299	298	299	299	299	299
Test (15°)	587 (C21, 9566, 9563)	196	582 (132, S7, 812)	195	274	274	274	273	274	274

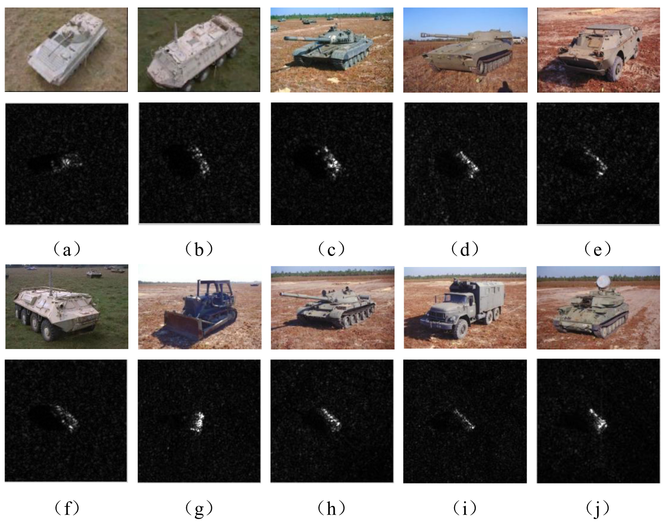


Fig. 7. Examples of ten types of ground targets. Optical images are on the top and SAR images are on the bottom. (a) BMP2. (b) BTR70. (c) T72. (d) 2S1. (e) BRDM2. (f) BTR60. (g) D7. (h) T62. (i) ZIL131. (j) ZSU234.

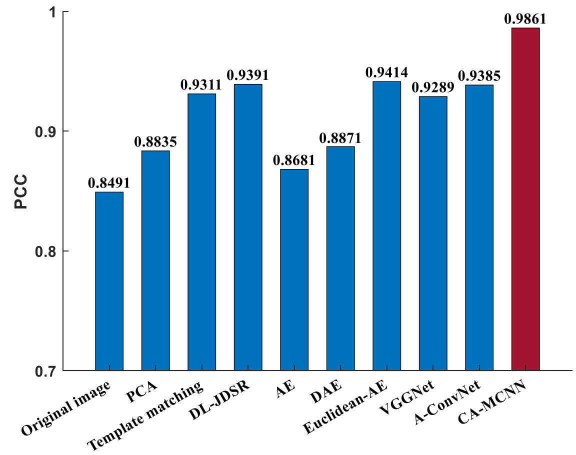


Fig. 8. Accuracies of different SAR target recognition methods on three-target data.

are 17° and 15°, respectively. Compared with the three-target data, the ten-target data also include the data from the other seven different ground targets. Similar to the three-target data, the depression angles of training data and test data are 17° and15°, respectively. The Caffe [23] deep learning framework is used to implement the experiments, which use a personal computer with Intel Xeon Silver 4114 CPU of 2.2 GHz and NVIDIA GTX 2080Ti GPU on Ubuntu 16.04 Linux system.

B. Three-Target MSTAR Data Experiments

In this section, we discuss the effectiveness of CA-MCNN on the three-target MSTAR data. First, the confusion matrix of CA-MCNN on three-target MSTAR data is shown in Table IV.

In the target recognition task, a widely used performance evaluation is the confusion matrix, each row of which is the actual category and each column of which is the predicted category. In addition, the elements of the confusion matrix denote the probabilities that the targets are recognized as a certain class and the elements on the diagonal represent the recognition accuracy of a certain class. As shown in Table IV, the recognition accuracy of BMP2 is 0.9813, the recognition of BTR70 is up to 0.9949, and the recognition accuracy of T72 reaches 0.9880. The result of the confusion matrix shows that the CA-MCNN has good recognition performance.

To further prove the effectiveness of CA-MCNN, we compare it with some traditional SAR target recognition methods and some deep learning-based SAR target recognition methods. The traditional SAR target recognition methods used here include using the amplitude features of the original SAR images which are obtained by performing a modulo operation on the complex SAR data, PCA [6], the template matching method, and the dictionary learning and joint dynamic sparse representation (DL-JDSR) [24]. The classifiers used for the amplitude features and PCA are support vector machine (SVM). The DL-JDSR using two kinds of features, namely the image domain amplitude feature and SIFT feature, and dictionary learning to recognize SAR targets. The deep learning-based SAR target recognition methods used here include the original auto-encoder (AE) [25], the denoising AE (DAE) [26], the Euclidean distance restricted AE [27] (Euclidean-AE for short), the VGG CNN (VGGNet) [19], and the A-ConvNet [1]. The classifiers used for AE, DAE, and Euclidean-AE are SVM. In particularly, because our proposed method does not need to expand the training data, to ensure fairness, we conduct comparison experiments without data augmentation based on the network structure and parameters provided in the relevant references. The intuitional recognition results of CA-MCNN and above-mentioned compared methods are shown in Fig. 8. The recognition results are represented by the probability of correct classification (PCC), which is calculated by dividing the number of correctly recognized targets by the total number of all targets. Table V lists the detailed recognition results of our method and the above-mentioned compared methods on three-target MSTAR data.

TABLE IV
CONFUSION MATRIX OF CA-MCNN ON THREE-TARGET MSTAR DATA

Type	BMP2	BTR70	T72
BMP2	0.9813	0.0034	0.0153
BTR70	0.0051	0.9949	0
T72	0.0052	0.0069	0.9880

Fig. 8 gives the PCC of all methods. It is clear that, compared with original image method, PCA method, template matching method, and DL-JDSR, CA-MCNN performs better on PCC. In addition, CA-MCNN also outperforms the compared deep learning methods, i.e., AE, DAE, Euclidean-AE, VGGNet, and A-ConvNet. Especially, we can see from Table V, for the BMP2 and the T72, the recognition results of CA-MCNN attain 0.9813 and 0.9880, respectively, which outperforms all other compared SAR ATR methods. The reason is there are some configuration variants in the test data of BMP2 and T72, and CA-MCNN combines the global information with the component information of the target to obtain a more efficient and robust feature representation. As a result, the performance of all other compared methods cannot rival the CA-MCNN. For the BTR70 which does not contain variants, the template matching method and the VGGNet can correctly recognize all test samples, at the same time, the accuracy of CA-MCNN is up to 0.9949, which is very close to 1. In addition, in terms of the PCC for three-target MSTAR data, CA-MCNN is at least 4.5% higher than other compared methods. The experimental results of three-target MSTAR data demonstrate the capability of CA-MCNN for SAR ATR and indicate that our proposed classification scheme can better make use of the global information and the component information to get a more efficient and robust feature representation of the target.

C. Ten-Target MSTAR Data Experiments

In this section, we evaluate the target recognition performance of CA-MCNN with the ten-target MSTAR data. Similar to Section IV-B, the confusion matrix of CA-MCNN on ten-target MSTAR data is shown first in Table VI. From Table VI, it is clearly that the accuracies of all target types, except T72 and ZSU234, are over 0.97. The best accuracy is shown in ZIL131, where all test samples are correctly classified. The accuracies on BTR60, 2S1, BRDM, D7, and T62 are close to 1. And even the worst accuracy is over 0.93.

Next, we compare the PCC of CA-MCNN with the PCCs of the above-mentioned traditional SAR target recognition methods and deep learning-based SAR target recognition methods. The results are shown in Fig. 9 and Table VII.

It can be seen form Fig. 9 and Table VII, CA-MCNN outperforms all the other compared SAR ATR methods. Especially, for the types of BMP2, BTR70, T72, BTR60, 2S1, D7, T62, and ZIL131, CA-MCNN yield the highest accuracy. For the BRDM type, the VGGNet has the highest accuracy and CA-MCNN follows closely with the accuracy up to 0.9927. For the ZSU234 type, the highest accuracy is shown

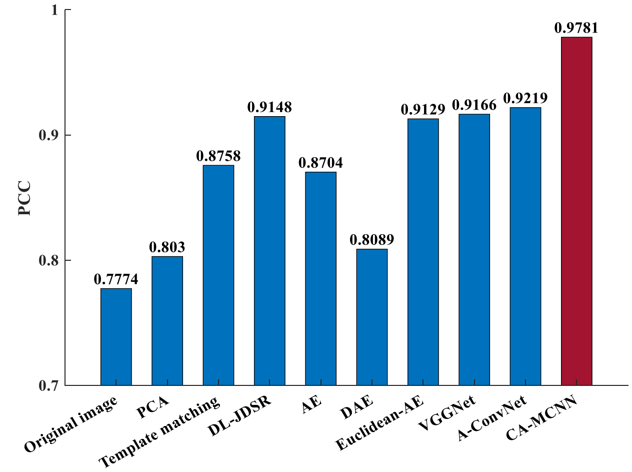


Fig. 9. Accuracies of different SAR target recognition methods on ten-target data.

in A-ConvNet, which correctly recognizes all test samples of ZSU234 type. And in terms of the PCC, CA-MCNN is at least 5.62% higher than other compared methods. The experimental results on ten-target MSTAR data further validate that CA-MCNN still has a better performance when the target types increase.

D. Model Analysis

1) Impact of ASC Components Fusion: As mentioned above, there are several strategies can achieve ASC components fusion. To understand the impact of different fusion strategies on the performance of SAR ATR, we show the recognition results on three-target MSTAR data when the ASC components fusion selects different fusion strategies in Table VIII.

From Table VIII, it can be seen that when the ASC components fusion select the joint function $\max(\cdot) + \text{mean}(\cdot)$, the whole network can obtain the highest PCC on the three-target MSTAR data. Therefore, we choose the joint function $\max(\cdot) + \text{mean}(\cdot)$ to achieve the ASC components fusion in the final version of CA-MCNN.

2) Ablation Study: Ablation study is always adopted to gain a better understanding of the network's behavior, which removes or replaces one or more certain components of the network to see how each component affects the performance. In our method, there are two main components that affect the performance of SAR ATR, i.e., the fusion of multiscale feature maps and the combination of global information and component information. Therefore, we designed several controlled experiments in this section to see how each component affects the performance. As for all controlled experiments, except for some certain examined components, the rest of setting remain consistent. Table IX summarizes the experiment results of ablation study on three-target MSTAR data. In Table IX, "backbone module" denotes the backbone module in CA-MCNN, which contains convolution layers and the fusion of multiscale feature maps. Compared with the "backbone module," "backbone module 2" does not have the fusion of multiscale feature maps, the rest of setting remain

TABLE V
DETAILED RECOGNITION RESULTS OF SEVERAL SAR TARGET RECOGNITION METHODS ON THREE-TARGET DATA

	BMP2	BTR70	T72	PCC
CA-MCNN	**0.9813**	0.9949	**0.9880**	**0.9861**
original image	0.7325	0.9643	0.9278	0.8491
PCA	0.8330	0.9541	0.9106	0.8835
Template matching	0.9148	**1**	0.9244	0.9311
DL-JDSR [25]	0.9301	0.9898	0.9312	0.9391
AE	0.8756	0.9439	0.8351	0.8681
DAE	0.7922	0.9796	0.9519	0.8871
Euclidean-AE [28]	0.9421	0.9388	0.9416	0.9414
VGGNet	0.8859	**1**	0.9485	0.9289
A-ConvNet	0.9199	0.9898	0.9399	0.9385

TABLE VI
CONFUSION MATRIX OF CA-MCNN ON TEN-TARGET MSTAR DATA

Type	BMP2	BTR70	T72	BTR60	2S1	BRDM	D7	T62	ZIL131	ZSU234
BMP2	0.9727	0.0017	0.0017	0.0034	0.0068	0	0	0	0	0.0136
BTR70	0.0051	0.9898	0	0.0051	0	0	0	0	0	0
T72	0.0051	0	0.9385	0	0	0	0	0.0308	0.0154	0.0103
BTR60	0	0	0	0.9964	0	0.0036	0	0	0	0
2S1	0.0036	0	0	0	0.9964	0	0	0	0	0
BRDM	0	0	0	0.0073	0	0.9927	0	0	0	0
D7	0	0	0	0	0	0	0.9963	0	0.0037	0
T62	0	0	0	0	0	0	0	0.9964	0.0036	0
ZIL131	0	0	0	0	0	0	0	0	1	0
ZSU234	0.0052	0.0017	0	0.0017	0	0	0.0481	0.0017	0	0.9416

TABLE VII
DETAILED RECOGNITION RESULTS OF SEVERAL SAR TARGET RECOGNITION METHODS ON TEN-TARGET DATA

	BMP2	BTR70	T72	BTR60	2S1	BRDM	D7	T62	ZIL131	ZSU234	PCC
CA-MCNN	**0.9727**	**0.9898**	**0.9385**	**0.9964**	**0.9964**	0.9927	**0.9963**	**0.9964**	**1**	0.9416	**0.9781**
Original image	0.6899	0.8673	0.7131	0.7897	0.4453	0.9307	0.8905	0.7802	0.9124	0.9562	0.7774
PCA	0.7070	0.8520	0.7715	0.8051	0.6971	0.7920	0.9598	0.8645	0.7956	0.9453	0.8030
Template matching	0.8637	0.9235	0.6993	0.9179	0.8577	0.8869	0.9818	0.9670	0.9307	0.9745	0.8758
DL-JDSR[25]	0.8876	0.9388	0.8625	0.8821	0.8905	0.9161	0.9854	0.9670	0.9234	0.9818	0.9148
AE	0.8245	0.9082	0.6735	0.8718	0.9161	0.9562	0.9708	0.9451	0.9234	0.9891	0.8704
DAE	0.7155	0.8980	0.7371	0.7436	0.5657	0.9599	0.9088	0.8498	0.9380	0.9672	0.8089
Euclidean AE [28]	0.8790	0.9286	0.7955	0.9179	0.9380	0.9672	0.9891	0.9414	0.9453	0.9964	0.9129
VGGNet	0.7683	0.9745	0.8872	0.9270	0.9818	**0.9964**	0.9780	0.9891	0.9854	0.8883	0.9166
A-ConvNet	0.8961	0.9745	0.7887	0.9641	0.9197	0.9818	0.9526	0.9597	0.9891	**1**	0.9219

consistent with "backbone module." "Global information" means the global feature representation, $S1$, mentioned above. "Component information" means the component feature representations, $S2$ and $S3$, mentioned above. From the rows 1 and 2 in Table IX, it can be seen that the PCC using only backbone module 2 and global information is 0.9568. And the PCC using only backbone module 2 and component information is 0.9407. Compared with the recognition results of other SAR ATR methods shown in Table V, the PCCs in rows 1 and 2 are higher, which demonstrates that the global information and component information used in our method are effective. In addition, when comparing the

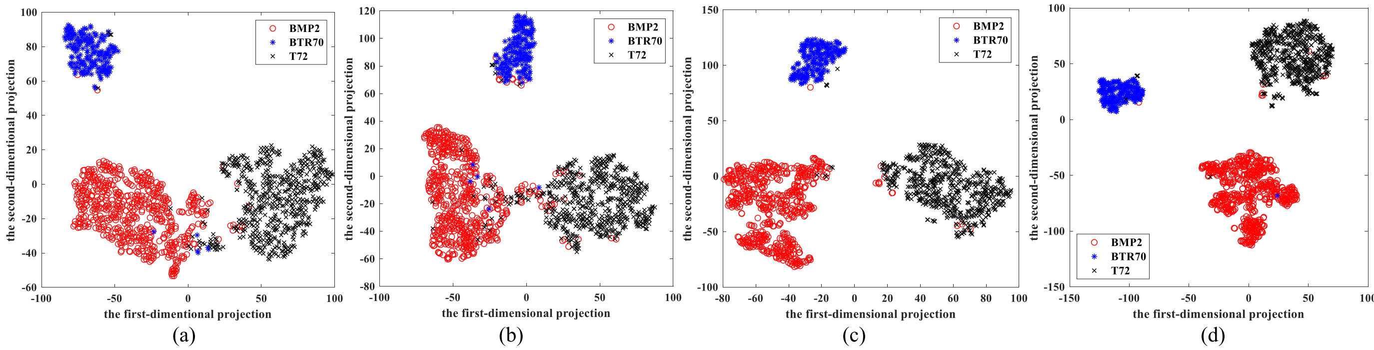


Fig. 10. t-SNE visualizations of the learned features. (a)–(d) Learned features corresponding to the four experimental settings in Table IX (each row corresponds to an experimental setting), respectively. For example, (a) exhibits the visualization result of the learned features obtained by the network corresponding to the first row in Table IX.

TABLE VIII
RECOGNITION RESULTS OF DIFFERENT FUSION STRATEGIES ON THREE-TARGET MSTAR DATA

Fusion Strategies	**PCC**
max(·)	98.17
mean(·)	98.32
max(·)+mean(·)	**98.61**
1_1C(cat(max(·),mean(·)))	98.39
attention	98.02

TABLE IX
EXPERIMENTAL RESULTS OF ABLATION STUDY ON THREE-TARGET MSTAR DATA

Components?				PCC
Backbone module	Backbone module2	Global information	Component information	
×	√	√	×	0.9568
×	√	×	√	0.9407
×	√	√	√	0.9788
√	×	√	√	**0.9861**

rows 1–3, we can see that the fusion of global information and component information is beneficial, which can obtain a more efficient feature representation of the target to achieve a better recognition performance. Furthermore, as shown in rows 3 and 4, whether the backbone module contains the fusion of multiscale feature maps will affect the PCC of SAR ATR. When the backbone module contains the fusion of multiscale feature maps, the PCC can be 0.73% higher than that without the fusion of multiscale feature maps, which demonstrates the effectiveness of the fusion of multiscale feature maps in the backbone module. The experimental results of ablation study on three-target MSTAR data demonstrate that the two components of CA-MCNN, i.e., the fusion of multiscale feature maps and the combination of global information and component information, are reasonable and effective.

3) Feature Analysis: Through comparisons with existing methods and detailed ablation studies, the quantitative performance analysis has proved the effectiveness of our method. In this section, we adopt the t-stochastic neighbor embedding (SNE) algorithm [28] to visualize the feature learned by CA-MCNN, as well the features learned through the other three experimental settings shown in Table IX. Fig. 10 exhibits the visualization results of the learned features corresponding to the four experimental settings (each row corresponds to an experimental setting) shown in Table IX on three-target MSTAR data. For example, Fig. 10(a) exhibits the visualization result of high-dimensional features obtained by the network corresponding to the first row in Table IX. It can be seen from the Fig. 10, compared with Fig. 10(a)–(c) has more obvious boundaries between all types, which shows the feature in Fig. 10(c) is more separable. In addition, compared with Fig. 10(c) and (d) shows a better feature distribution, in which each class gathers more closely and the margin between them is much more distinct. Besides, the number of misclassified points in each category in Fig. 10(c) is less than that in Fig. 10(b).

V. CONCLUSION

An SAR ATR method combining the global information with component information is discussed in this article. For traditional CNN-based methods, an important problem is that the component information of a target cannot be fully utilized. To learn the component information, we use the ASCs extracted from the target echoes as the components of the target. In our method, global information from the whole SAR image and component information corresponding to ASCs of the target are combined within CNN framework to learn a more efficient and robust target feature representation, which can better distinguish different classes of targets. In addition, to enhance the feature description ability, we fuse the multiscale feature maps of different layers. The experiments on the MSTAR data set show that the SAR ATR method proposed in this article has a higher PCC than comparative methods, which demonstrates the capability of our method for SAR ATR.

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